Active Maintenance Report: galos_coppock_2023

Galos and Coppock (2023, Science Advances): Gender Composition Predicts Gender Bias

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This repository holds the actively maintained replication code for Galos and Coppock (2023), together with the reproducibility report that documents what the original archive did and did

not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

Article	10.1126/sciadv.ade7979
Replication archive	10.7910/DVN/5MCRNY

The data are not redistributed here. The deposit is 19 files at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file against `original_manifest.csv`, which pins the file identifiers, folder paths, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not. The same script stops if `original/` holds anything the manifest does not list, since a stray file makes a working copy differ from a fresh clone in exactly the way a reproduction check exists to catch.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `galos_coppock_2023.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 19 files, produces every table and figure into `maintained/output/`, and rebuilds the number-by-number comparison table in `ground_truth/`. Required packages: tidyverse, metafor, broom, scales, rsample, modelsummary, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about two and a half minutes, almost all of it in the 500-resample bootstrap behind one standard error in the Results section.

A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.** The CSV, TeX and PNG files come back byte-identical. The eight figure PDFs always differ, because a PDF records the time it was written, so a changed hash there is not evidence of anything.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Almost. Eight of the ten deposited scripts run to completion on a current R installation with no changes at all. One fails outright and one runs but does not produce the published number.

`appendix_figures_and_loop.R` stops with an error on its first study. It binds a study's occupation-level estimates to a meta-regression row obtained from `broom::tidy()`, which carries no `fraction_female` column, and then sorts the combined frame with `fct_reorder(occupation_label, fraction_female)`. The meta row's ordering value is `NA`, and `forcats::lvls_reorder()` refuses to order a factor with a missing sort key. Because the error comes on the first iteration, the loop never reaches its last line, which is the one that writes `data/gender_gradients_df.rds`. That file is deposited and is correct, and re-deriving it from the occupation CATEs agrees with it to 4e-13 on every estimate and standard error, so nothing downstream is wrong. What is missing is any signal that the script and the object it is meant to write have come apart.

`analyze_cates.R` runs, but ships with `sims <- 5` where the paper used 500, and a comment saying so. Five resamples give a bootstrapped standard error of 0.54; five hundred give 0.578, which is the published 0.58. A reader who runs the archive as deposited gets a number that disagrees with the paper in the second decimal place, with no error and no warning.

Neither problem needs author intervention: the first is a two-line fix and the second is a one-character one. Every package the archive names still installs from CRAN.

Running the archive is not the same as reproducing the paper, and this archive is a clear case of the difference. The scripts cover Tables 1 and 2, Figs. 4 and 5, and five of the supplementary figures, though every one of them draws to the active graphics device and none calls `ggsave()`, so the deposit as shipped writes no figure files at all. They do not cover the 57 study-by-study panels of section S4, because the loop that draws them fails and its `ggsave()` calls are commented out in any case, and they do not cover the study manifest of table S2.1, for which the deposit contains no code. That is 57 published figures and one published table with no deposited source.

1.2 Does the maintained rewrite reproduce the paper?

The ground truth records 483 values. 450 of the 476 that can be checked against the article reproduce, including all 57 study-level average treatment effects and their standard errors from fig. S1.1, all 12 survey experiment effects from fig. S3.4, all 37 study-level gender gradients

from fig. S5.62, every coefficient, standard error, tau-squared and sample size in Tables 1 and 2, and all 335 estimate-and-standard-error labels printed on the 57 study panels.

The 26 that do not divide into two groups: 3 where the article disagrees with itself, and 23 where the deposit does not carry the quantity the published figure prints.

The article against itself. The paper reports that bias is larger in women-dominated than in men-dominated occupations "by a statistically significant 2.10 points". The archive and the rewrite both give 2.10547, which rounds to 2.11 at the printed precision; the published figure is the truncation. Published Table 2 prints the Model 2 intercept as −3.72 with no significance stars, where the coefficient's p-value is 0.001 and the numerically identical −3.73 intercept in Model 3 does carry its two stars; the estimate and its standard error are right and only the marker is missing. Table S2.1 gives 18 occupations for Gonzalez et al. and 24 for Dahl and Krog, where the corresponding study panels, figs. S4.37 and S4.32, plot 17 and 21.

The deposit against the figures. Every study panel labels each occupation with three lines: its name, the share of women in it, and a sample size. The first two reproduce for all 57 panels. The third is only partly recoverable, because the deposit carries no subject counts. What it does carry is the residual degrees of freedom of each occupation's own regression, and `df + 1` equals the published sample size exactly for 167 of the 298 occupations, which is every occupation of 35 of the 57 studies. For the other 22 the published figure is several times larger, which is what a design with several applications per job opening gives, and the deposit provides no way to recover the multiplier. The same gap shows up in table S2.1, whose subject counts reconstruct for the same 35 studies.

Not one of the 26 is a defect of the rewrite.

2 Paper overview

Citation: Galos, D. R. and Coppock, A. (2023). "Gender composition predicts gender bias: A meta-reanalysis of hiring discrimination audit experiments." *Science Advances*, 9, eade7979. DOI: 10.1126/sciadv.ade7979

Summary: Since 1983, more than 70 employment audit experiments have randomized the gender of fictitious job applicants, and the results point in both directions. This paper reconciles them with what the authors call a meta-reanalysis. The microdata for 57 studies reported in 51 papers are obtained or reconstructed, each study is reanalyzed to give conditional average treatment effects at the occupation level, occupational gender composition is merged in from International Labour Organization statistics, and the resulting 298 occupation-level estimates are meta-regressed on the share of women in the occupation. That slope is the gender gradient, and it is strongly positive: 8.15 percentage points per unit share female, with a standard error of 1.50. In a fully men-dominated occupation the expected effect of being a woman is −3.02 points; in a fully women-dominated one it is +5.13. Discrimination, in other words, preserves the status quo gender distribution rather than correcting it. The gradient survives country,

year, and study fixed effects, holds separately among majority and minority status applicants, and reappears in a parallel meta-analysis of 12 vignette survey experiments.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0.

Script	Status on current R	Resolution
helpers.R	Clean (sourced by every analysis script)	No changes required
analyze_ates.R	Clean	No changes required
analyze_cates.R	Runs, but at 5 bootstrap resamples rather than the published 500	Set sims to 500, as the script's own comment instructs
analyze_cates_ethnicity.R	Clean	No changes required
analyze_cates_regression_tables.R	Runs; warns that redundant predictors were dropped from one model	No changes required; the warning is about collinear fixed effects
appendix_cates_survey_field.R	Clean	No changes required
appendix_evidence_map.R	Runs; warns that the size aesthetic for lines is deprecated	Use linewidth instead of size
appendix_figures_and_loop.R	Fails at fct_reorder on an NA ordering value	Replace the NA sort key before reordering
appendix_gender_gradient.R	Clean; reads the data file the failing loop script is meant to write	No changes required
appendix_survey_ates.R	Clean	No changes required

Every package the archive's README names still installs from CRAN: `tidyverse`, `metafor`, `broom`, `scales`, `texreg` and `xtable`. `texreg` is the one worth checking rather than assuming, because it is a table-formatting package that writes only to the console; version 1.39.5 was released in December 2025 and is actively maintained. `xtable` is loaded by `appendix_figures_and_loop.R` and never called.

The archive's own checks are reproduced by `ground_truth/archive_values.R`, which sources the deposited scripts unmodified, harvests the objects they assign and the LaTeX tables they print, and writes `ground_truth/archive_values.csv`. That file is the source of the `value_script` column below. It is not part of `run_all.R`, because "does the deposit reproduce the paper" and "does the rewrite reproduce the paper" are different questions and answering the first one should not be a precondition for running the second.

3.1 The loop that cannot write its own output

`appendix_figures_and_loop.R` walks the 57 studies, fits a within-study meta-regression wherever a study distinguishes three or more occupations, and draws one appendix figure per study. Inside the loop it builds a plotting frame:

```
local_dat <- local_dat |> bind_rows(local_meta)
local_dat <- local_dat |>
  ungroup() |>
  mutate(occupation_label = fct_reorder(occupation_label, fraction_female))
```

`local_meta` comes from `tidy()` applied to an `rma.uni` fit, and a tidied model has no `fraction_female` column, so `bind_rows()` fills that cell with `NA`. `fct_reorder()` then asks `lvls_reorder()` to order a factor by a vector containing a missing value, which errors. Replacing the missing key with `-Inf` before reordering fixes it and puts the meta row first, which is where the published appendix figures show it.

The loop's `ggsave()` calls are commented out in the deposit, so the only thing it produces that anything else consumes is the last line, `write_rds(gender_gradients_df, "data/gender_gradients_df.rds")`. That file is deposited, and `appendix_gender_gradient.R` reads it to build fig. S5.62. The archive is therefore internally consistent while being unable to rebuild itself: the deposited object is right, and the code that made it does not run.

That last line is also a hazard for anyone checking the archive, because it writes into the archive directory and would replace a deposited file with the output of the run. Deposited code is run in a scratch copy under `tempdir()` here for that reason, and `original/` is verified against the manifest on every run.

3.2 The bootstrap that ships at five iterations

`analyze_cates.R` estimates the difference in the magnitude of bias between fully women-dominated and fully men-dominated occupations, and bootstraps its standard error:

```
set.seed(343)
# sims <- 500
sims <- 5
```

The published standard error is 0.58. At five resamples the archive prints 0.54.

4 Number-by-number comparison

The ground truth is at `ground_truth/galos_coppock_2023_ground_truth.csv` and is generated by `ground_truth/build_ground_truth.R`, which runs as the last step of `run_all.R`. Its only typed numbers are the `value_paper` column, read from the published article and its supplementary materials; `value_script` is read from the harvest of the deposited code and `value_rewrite` out of `maintained/output/`. The verdicts are computed at the precision the article prints rather than asserted.

It records 483 values. 300 of them have both a published counterpart and a deposited-code counterpart, and of those 295 match and 5 do not.

Most of the published values are inside figures rather than in prose. Figures S1.1, S3.4 and S5.62 print the estimate and standard error beside every point, which makes 218 published numbers that no sentence in the paper states. The 57 study panels of section S4 print another 335. All of them are extracted from the supplementary PDF and checked one by one.

Table 2: Where the checkable published numbers are, and how the rewrite does against them.

Location	Published values checked	Rewrite matches	Rewrite mismatches
figs_S4.5_to_S4.61	173	151	22
fig_S1.1	116	116	0
fig_S5.62	76	76	0
fig_S3.4	26	26	0
table_2_model3	14	14	0
table_1_model1	8	8	0
table_1_model2	8	8	0
table_1_model3	8	8	0
table_1_model4	8	8	0
table_1_model5	8	8	0
table_2_model1	8	8	0
table_2_model2	8	7	1
text	8	7	1
table_S2.1	4	2	2
text_S5	2	2	0
text_S3	1	1	0

Table 3: Every published value the rewrite does not match, and where the fault lies.

Location	Quantity	Paper	Rewrite	Where the fault lies
text	Difference in magnitude, 100 vs 0 percent women (pp)	2.1	2.105	paper_internal
table_2_model2	Intercept significance stars (0 = none, 1 = one, 2 = two, 3 = three)	0.0	2.000	paper_internal
table_S2.1	Subject counts reproduced	57.0	35.000	archive
table_S2.1	Occupation counts reproduced	57.0	55.000	paper_internal
fig_S4.8	Occupation sample size labels, Alden et al. (68): Sweden	2.0	0.000	archive
fig_S4.11	Occupation sample size labels, Asali et al. (70): Georgia	9.0	0.000	archive
fig_S4.13	Occupation sample size labels, Baert et al. (71): Belgium	1.0	0.000	archive
fig_S4.14	Occupation sample size labels, Baert et al. (72): Belgium	1.0	0.000	archive
fig_S4.17	Occupation sample size labels, Berson (75): France	1.0	0.000	archive
fig_S4.28	Occupation sample size labels, Carlsson (13): Sweden	13.0	0.000	archive
fig_S4.32	Occupation sample size labels, Dahl and Krog (84): Denmark	21.0	3.000	archive
fig_S4.33	Occupation sample size labels, Edo et al. (85): France	3.0	0.000	archive
fig_S4.37	Occupation sample size labels, Gonzalez et al. (89): Spain	17.0	0.000	archive
fig_S4.38	Occupation sample size labels, Hipp (90): Germany	1.0	0.000	archive
fig_S4.42	Occupation sample size labels, Neumark et al. (9): United States	1.0	0.000	archive
fig_S4.43	Occupation sample size labels, Nunley et al. (93): United States	6.0	0.000	archive
fig_S4.44	Occupation sample size labels, Oreopoulos (94): Canada	21.0	1.000	archive
fig_S4.46	Occupation sample size labels, Pedulla (96): United States, field experiment	4.0	0.000	archive
fig_S4.47	Occupation sample size labels, Petit (97): France	1.0	0.000	archive
fig_S4.50	Occupation sample size labels, Riach and Rich (8): Australia	7.0	0.000	archive

Table 3: Every published value the rewrite does not match, and where the fault lies. *(continued)*

Location	Quantity	Paper	Rewrite	Where the fault lies
fig_S4.51	Occupation sample size labels, Riach and Rich (100): England	4.0	0.000	archive
fig_S4.52	Occupation sample size labels, Rivera and Tilcsik (101): United States, field experiment	1.0	0.000	archive
fig_S4.53	Occupation sample size labels, Rooth (102): Sweden	7.0	0.000	archive
fig_S4.54	Occupation sample size labels, Ruffle and Shtudiner (103): Israel	10.0	1.000	archive
fig_S4.60	Occupation sample size labels, Zhou et al. (108): China	4.0	0.000	archive
fig_S4.61	Occupation sample size labels, Capéau et al. (82): Belgium	1.0	0.000	archive

The 7 remaining recorded quantities are ones the scripts produce and the paper never prints: the mean and median number of occupations per study, the count of survey occupation CATEs, and the intercept and slope of the two fitted lines in fig. S3.3, which are drawn but never stated numerically. They are recorded with no published counterpart rather than counted as matches.

Four published floats have no row here, and none of them has any source in the deposit. Figures 1, 2 and 3 of the article are conceptual diagrams and a coding summary rather than analyses of the meta-reanalysis data. Figure S6.63 is a PRISMA flow diagram recording the literature search, whose counts come from the screening process and not from any data file in the deposit.

5 Tables 1 and 2

Both published tables reproduce in full, coefficients, standard errors, significance markers, tau-squared and sample sizes alike, with the single exception of the Model 2 intercept marker in Table 2.

Table 4: Table 1 as reproduced by the maintained rewrite. Standard errors in parentheses.

Model	Intercept	Gender gradient	Tau squared	Nobs
Model 1	-3.02 (0.81)	8.15 (1.50)	12.57	298
Model 2	-2.48 (1.58)	8.33 (1.58)	11.71	298
Model 3	-5.43 (1.91)	8.69 (1.56)	12.56	298
Model 4	-5.20 (1.82)	7.97 (1.57)	10.22	298
Model 5	1.05 (1.96)	7.51 (1.54)	8.89	298

Table 5: Table 2 as reproduced by the maintained rewrite. Model 1 is the majority status group, model 2 the minority status group, model 3 both. Standard errors in parentheses.

term	Model 1	Model 2	Model 3
Intercept	-1.55 (1.31)	-3.72 (1.13)	-3.73 (1.14)

Gender gradient	5.70 (2.50)	9.08 (2.26)	9.11 (2.28)
Majority status	NA	NA	2.19 (1.74)
Majority x gradient	NA	NA	-3.42 (3.37)

6 Figures

The main text carries two figures of results and the supplementary materials sixty-one more, of which the rewrite reproduces all sixty-one: the two forest plots, the evidence map, the survey-against-field comparison, the gender gradient forest plot, and the 57 study-by-study panels.

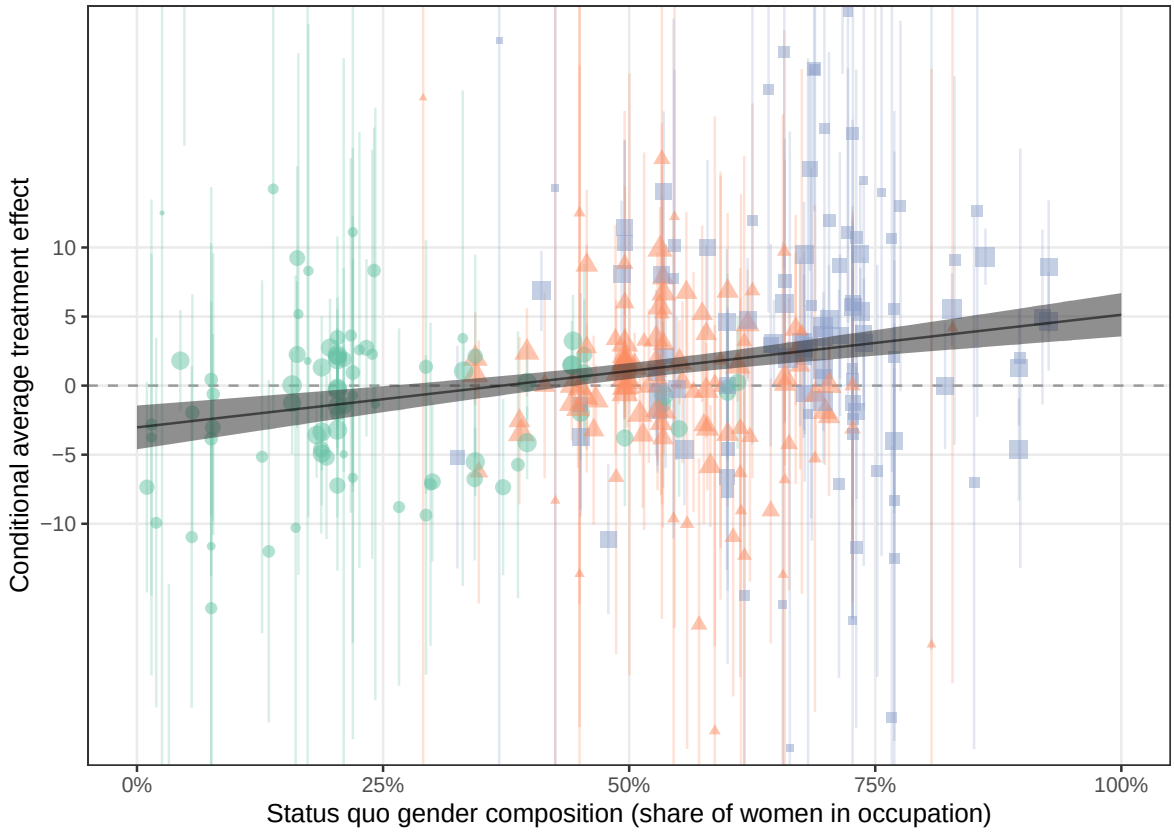


Figure 1: Figure 4: occupation-level CATEs against gender composition, with the Table 1 model 1 regression line.

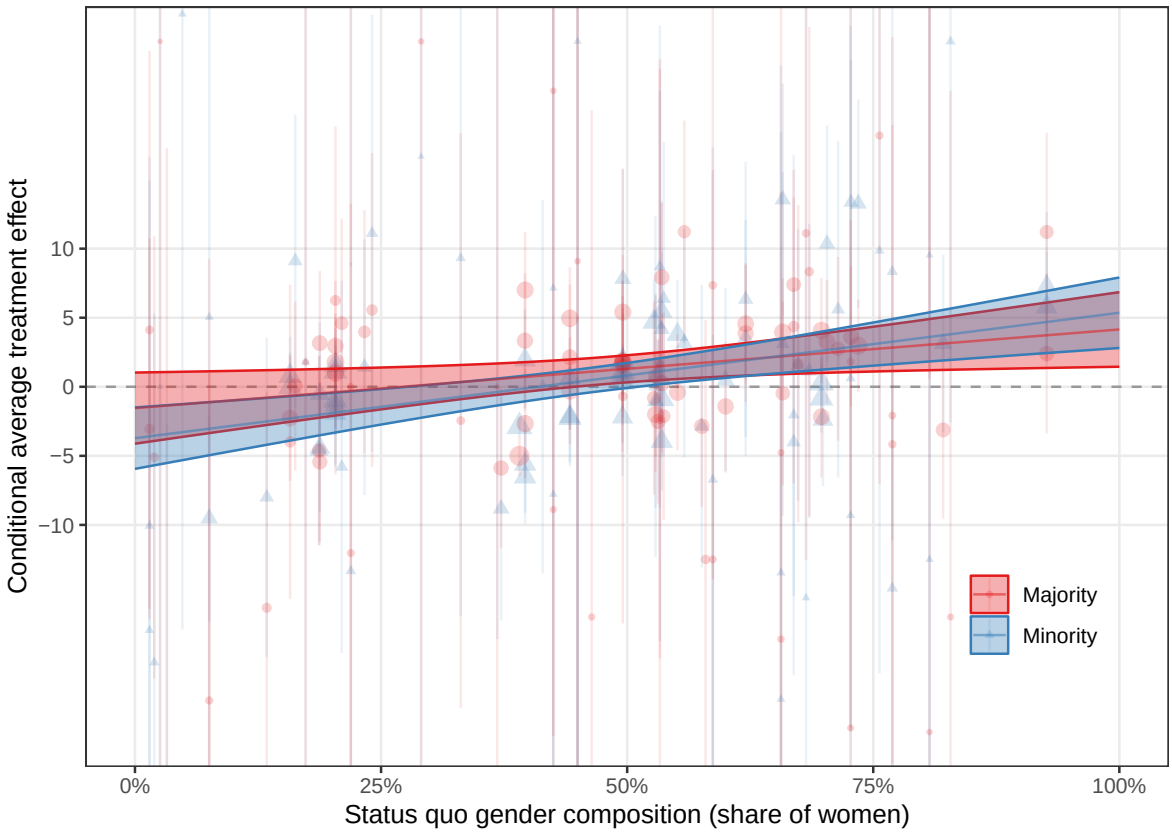


Figure 2: Figure 5: the same estimates split by majority and minority applicant status.

6.1 Figure S1.1

Fig. S1.1 is a forest plot of all 57 field studies, each labelled with its estimate and standard error. All 116 of those labels reproduce, along with the meta-analytic summary of 1.2 points with a standard error of 0.4.

Table 6: The three largest and three smallest of the 57 study-level effects in fig. S1.1, all of which reproduce.

Study	ATE (SE)
Albert et al. (67): Spain	9.9 (1.8)
Birkelund et al. (76): the Netherlands	8.3 (3.2)
Becker et al. (74): Austria, Germany, Switzerland	8.1 (0.9)
Rivera and Tilcsik (101): United States, field experiment	-3.8 (3.3)
Arai et al. (69): Sweden (equivalent CVs)	-6.3 (4.0)
Neumark et al. (9): United States	-12.3 (8.4)

6.2 Figures S4.5 to S4.61

The 57 study panels are the densest published numbers in the paper and the part of it the deposit does not draw. `figure_s4_study_panels.R` rebuilds them as one paginated PDF, one page per study, and writes every plotted value to CSV. All 335 estimate-and-standard-error labels and all 298 gender composition labels reproduce. The sample size labels are the exception described above: `df + 1` recovers 167 of the 298. That is every occupation of 35 studies, plus five scattered across the other 22.

6.3 Figure S5.62

The 37 study-level gender gradients are the paper's own evidence that a single audit experiment cannot see the pattern it documents: only 7 of the 37 are individually significant, which the paper glosses as putting the power of the average study at roughly 19 percent. All 37 estimates, their standard errors, and the meta-analytic 9.0 with a standard error of 2.1 reproduce.

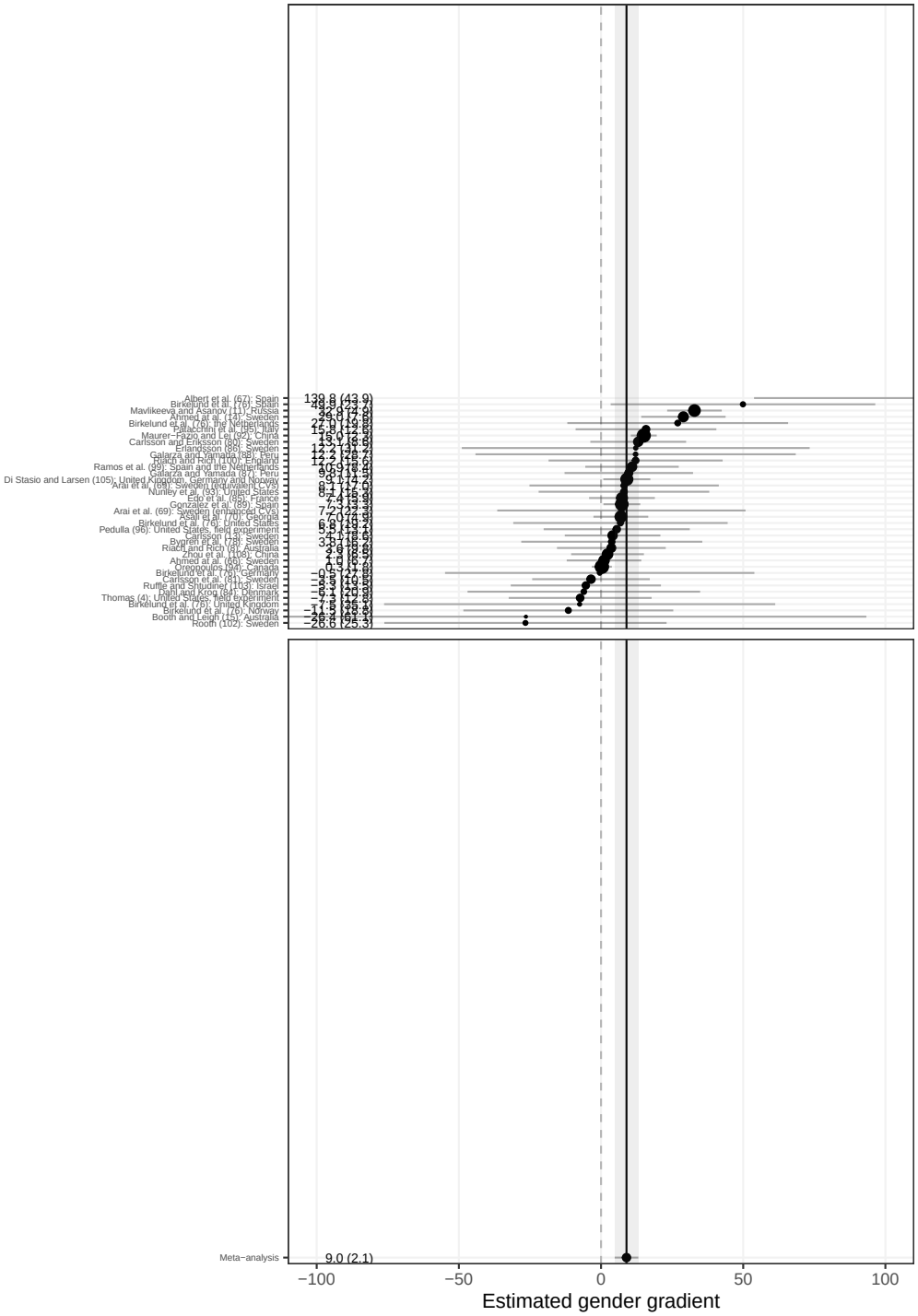


Figure 3: Figure S5.62 as reproduced by the maintained rewrite.

7 Maintained rewrite

The rewrite lives in `maintained/`: `helpers.R` plus fourteen scripts covering one cleaning step, three published tables, eight figures, and two sets of in-text quantities. It is a translation, not a reanalysis. Every estimator, specification and sample restriction is the one the paper used, and the published tables come out of `metafor::rma.uni()` with the same formulas the archive writes.

7.1 Architecture

Table 7: The maintained rewrite.

Script	Produces
helpers.R	Packages, paths, formatting helpers
clean_gender_gradients.R	gender_gradients.csv
table_1_meta_regression.R	table_1_meta_regression.tex and .csv
table_2_ethnicity_meta_regression.R	table_2_ethnicity_meta_regression.tex and .csv
table_s2_1_study_manifest.R	table_s2_1_study_manifest.tex and .csv
figure_4_cates_gender_gradient.R	Figure 4
figure_5_cates_ethnicity.R	Figure 5
figure_s1_1_field_ates_forest.R	Figure S1.1 and its 116 plotted values
figure_s2_2_evidence_map.R	Figure S2.2
figure_s3_3_survey_field_comparison.R	Figure S3.3 and the two fitted gradients
figure_s3_4_survey_ates_forest.R	Figure S3.4 and its 26 plotted values
figure_s4_study_panels.R	Figures S4.5 to S4.61 and their 335 plotted values
figure_s5_62_gender_gradient_forest.R	Figure S5.62 and its 76 plotted values
text_summary_stats.R	The counts quoted in the text and in section S5
text_difference_in_magnitude.R	The predicted CATEs and the bootstrapped SE

Four decisions are worth stating.

The study-level gender gradients are recomputed, not read. `clean_gender_gradients.R` refits the 37 within-study meta-regressions from the occupation CATEs rather than loading the deposited `gender_gradients_df.rds`. Depending on that file would mean depending on an intermediate the deposited code cannot regenerate. The two agree to 4e-13.

The uncovered floats are covered. `figure_s4_study_panels.R` and `table_s2_1_study_manifest.R` rebuild the 57 study panels and the study manifest, neither of which the deposit produces. Rebuilding them is what makes the gaps in the deposit measurable: without a reconstruction there is nothing for the 335 panel labels and the 171 manifest cells to be compared against.

Every value that appears in `output/` is computed from the deposited data. No published number is typed into a script as an input to a calculation, and the figures write their plotted labels to CSV so that a value shown in a figure can be traced to the fit that produced it rather than to a transcription.

The bootstrap runs at 500 resamples. That is the cost of the published standard error and it is paid rather than avoided. `rsample::bootstraps()` at `set.seed(343)` gives the same resamples as the archive's `sample_n()` loop, so the two agree exactly at 0.5781.

7.2 Patterns replaced

Table 8: Patterns replaced in the maintained rewrite.

Original pattern	Replacement	Reason
magrittr pipe	native pipe	House style
‘rm(list = ls())‘	(omitted)	House style
relative paths from the archive root	‘here::here()‘	Scripts must run from the project root
‘texreg()‘ printed to the console	‘modelsummary(output = ...)‘ to ‘.tex‘, plus ‘write_csv()‘	Console-only output cannot be diffed between runs
‘size =‘ in ‘geom_hline()‘	‘linewidth =‘	Deprecated in ggplot2 3.4.0; warns on every run
‘fct_reorder()‘ on an NA sort key	replace the key with ‘-Inf‘ before reordering	Errors in forcats 1.0.0
‘legend.position = c(x, y)‘	‘legend.position = ”inside”‘ with ‘legend.position.inside‘	House style; the numeric form still works
‘sims <- 5‘ with a comment to raise it	‘sims <- 500‘	Recovers the published standard error
manual ‘sample_n()‘ bootstrap loop	‘rsample::bootstraps()‘	Same resamples at the same seed, in a maintained package
‘predict.rma()‘ called directly	‘predict()‘ with S3 dispatch	Method called by name rather than through the generic

`xtable` is loaded by the archive but never called, so it is simply dropped. `texreg` still installs and still works; it is replaced because it writes only to the console, which means the published tables exist nowhere as files and cannot be diffed between runs.

8 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23

Table 10: Package versions used for the run behind this report.

Package	Version
tidyverse	2.0.0

metafor	5.0.1
broom	1.0.13
scales	1.4.0
rsample	1.3.2
modelsummary	2.6.0
knitr	1.51
kableExtra	1.4.0
here	1.0.2